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Project Title :	Dynamic Portfolio Risk Arbitrage & Value-at-Risk (VaR) Monte Carlo Simulator
Week :	Week 5

Week 5 – Business Analytics

1. Objective

Develop a financial risk assessment platform that evaluates portfolio performance under changing market environments. The system aims to estimate investment risk, optimize asset allocation, and forecast portfolio outcomes using statistical techniques.

Goals

- Analyze portfolio performance metrics.
- Evaluate asset diversification effectiveness.
- Predict future portfolio returns.
- Measure investment risk exposure.
- Generate portfolio optimization recommendations.
- Support strategic financial decision-making.

2. Input Data

The system requires:

- Asset names and categories
- Historical monthly returns
- Investment amounts
- Benchmark market index data
- Risk-free interest rate
- Portfolio weight distribution
- Forecast horizon

3. Asset Performance Analysis

Calculate asset growth rates using:

Growth Rate = ((Final Value − Initial Value) / Initial Value) × 100

Example:

Initial Value = ₹500

Final Value = ₹575

Growth Rate = ((575 − 500) / 500) × 100

= 15%

4. Risk Metrics

The platform computes:

- Mean Return
- Variance
- Standard Deviation
- Sharpe Ratio
- Beta Value
- Maximum Drawdown

These indicators help investors understand portfolio stability and performance.

5. Portfolio Diversification Analysis

The diversification module examines:

- Asset concentration levels
- Sector exposure
- Geographic distribution
- Correlation between assets

Benefits:

- Reduces overall risk
- Improves return consistency
- Enhances investment balance

6. Risk-Adjusted Performance

Sharpe Ratio

Sharpe Ratio = (Portfolio Return − Risk-Free Rate) / Portfolio Volatility

Higher values indicate better risk-adjusted returns.

7. Scenario-Based Market Testing

The system generates multiple market situations:

- Bull Market
- Bear Market
- Sideways Market
- High Volatility Market

Each scenario estimates possible portfolio outcomes.

8. Portfolio Optimization

Optimization identifies the best asset allocation by:

- Maximizing expected returns
- Minimizing risk
- Maintaining diversification

Techniques Used:

- Mean-Variance Optimization
- Efficient Frontier Analysis

9. Forecasting Future Returns

Future portfolio performance is projected using:

- Historical return trends
- Moving averages
- Statistical forecasting models

Outputs include:

- Expected return
- Best-case outcome
- Worst-case outcome

10. Investment Decision Support

The system provides:

- Portfolio health score
- Risk category classification
- Rebalancing suggestions
- Asset allocation recommendations

11. Risk Classification

Risk Score	Category
0–20	Low Risk
21–50	Moderate Risk
51–80	High Ris

12. System Flow

Investment Data

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Data Validation

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Performance Analysis

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Risk Measurement

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Diversification Evaluation

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Scenario Testing

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Portfolio Optimization

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Forecast Generation

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Decision Support Report

13. Expected Output

1. Portfolio return summary

2. Risk analysis report
3. Diversification score
4. Sharpe ratio calculation
5. Forecasted portfolio value
6. Asset allocation chart
7. Risk classification result
8. Performance dashboard
9. Optimization recommendations
10. Investment insights

14. Technologies You Can Use

Programming Language: Python

Libraries:

- NumPy
- Pandas
- SciPy
- Matplotlib
- Plotly
- Scikit-Learn

15. Key Learning Outcomes

After completing this task, students will understand:

- Portfolio management concepts
- Investment risk analysis
- Asset diversification strategies
- Financial forecasting
- Portfolio optimization methods
- Performance measurement techniques
- Data-driven investment decisions
- Python-based financial analytics